a Cohort construction and patient-disjoint partitions

ED-linked parent
81,519 studies / 47,472 patients

→ **Reference-available core**
81,084 studies

Exclusions: 148 absent images;
286 unavailable diagnosis records;
1 absent report-source row.

One study per fitting/tuning patient, selected before report eligibility; test retains repeat studies.

FITTING POOL
37,827 patients / 2,264 coded +

COMMON TUNING
4,726 patients / 276 coded +

FIXED TEST
8,972 studies / 4,731 patients / 686 coded +

b Two records, different roles

Report state S controls eligibility

Explicit + / explicit - / uncertain / blank

Coded outcome Y stays unchanged

Same-stay pneumonia-related code, including aspiration

c Controlled supervision: N = 512, 1,536 or 3,072 patients per fit

Explicit reports

Eligibility: S = 0 or 1
Candidate pool: 3,937 patients

All states: random

Same fitting size N
Coded class counts may change

All states: class-matched

Same N and exact coded class counts
Membership within each class changes

Logistic: 20 draws; trees: 10 at N = 3,072. Unique patients within fits; source pools may overlap.

d Shared modeling information, paired evaluation

Two frozen image representations
+ recorded triage values
Common training-only transforms

→ Image / triage / fused heads
Common tuning panel and rules
Logistic + bounded tree sensitivity

→ **Fixed studies and coded reference**
Absolute fusion performance
Fusion increment = fusion - image

Post-hoc controls: admit blank or uncertain states separately; cross positive and negative source memberships.